Unit - IV
Tree

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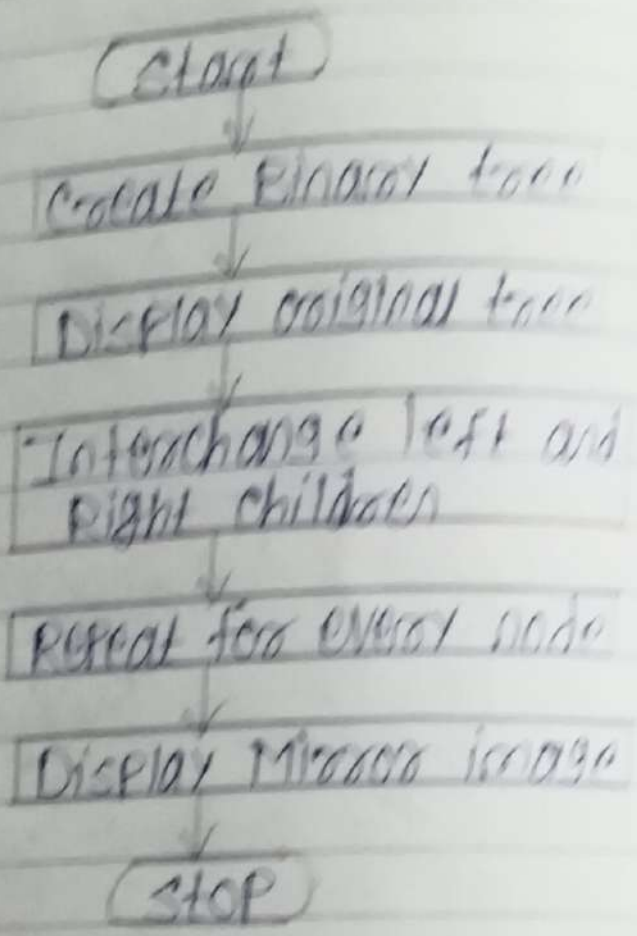
Title: Mirror image of Binary Tree

1) Problem statement:
To write a C++ Program to generate the mirror image of a binary tree in interchanging the left and right children of every node.

2) objectives:
i) To understand the concept of a binary tree.
ii) To create a binary tree using C++.
iii) To interchange the left and right children of every node.
iv) To display the original and mirror image of the binary tree.

3) Algorithm:
i) start
ii) Create a binary tree by inserting nodes.
iii) Display the original tree using inorder traversal.
iv) Call the mirror function.
v) Interchange the left and right children of each no
vi) Repeat the Process for the left and right subtrees.
vii) Display the mirror image using inorder traversal.
viii) stop.

4) Flowchart:


Start
Create Binary tree
Display original tree
Interchange left and right children
Repeat for every node
Display Mirrored image
Stop

```
#include <iostream>
using namespace std;

struct Node {
  int data;
  Node *left, *right;

  Node(int x) {
    data = x;
    left = right = NULL;
  }
};

Node* createTree() {
  int x;
  cin >> x;

  if (x == -1)
    return NULL;

  Node* root = new Node(x);

  root->left = createTree();
  root->right = createTree();

  return root;
}

void mirror(Node* root) {
  if (root == NULL)
    return;

  Node* temp = root->left;
```

```
        root->left = root->right;
        root->right = temp;

        mirror(root->left);
        mirror(root->right);
    }

    void inorder(Node* root) {
        if (root != NULL) {
            inorder(root->left);
            cout << root->data << " ";
            inorder(root->right);
        }
    }

    int main() {
        cout << "Enter tree nodes (-1 for NULL):\n";

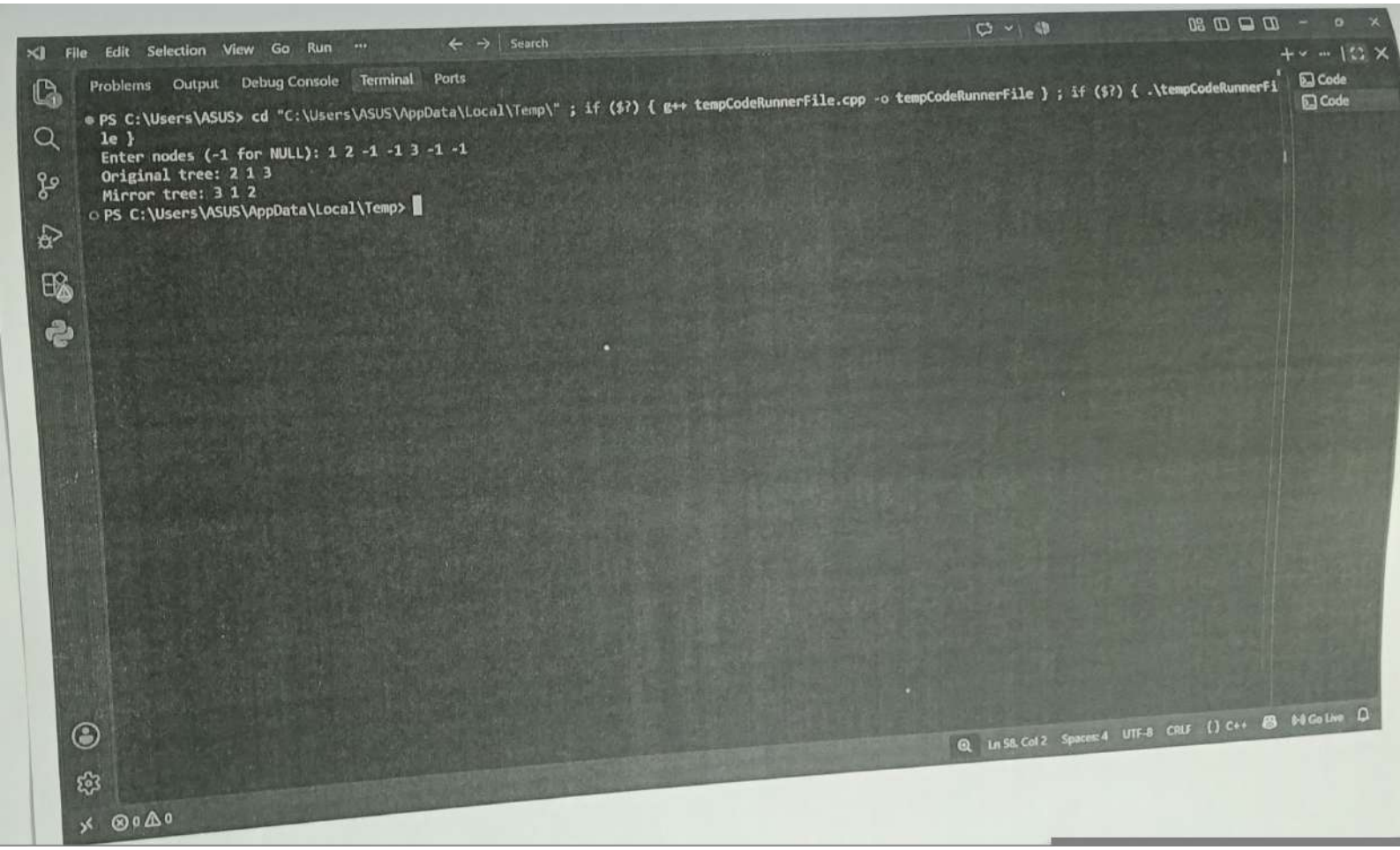
        Node* root = createTree();

        cout << "Original tree: ";
        inorder(root);

        mirror(root);

        cout << "\nMirror image: ";
        inorder(root);

        return 0;
    }
```


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Problems Output Debug Console Terminal Ports
PS C:\Users\ASUS> cd "C:\Users\ASUS\AppData\Local\Temp\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFi
le }
Enter nodes (-1 for NULL): 1 2 -1 -1 3 -1 -1
Original tree: 2 1 3
Mirror tree: 3 1 2
PS C:\Users\ASUS\AppData\Local\Temp>
Code
Code
Ln 58, Col 2 Spaces: 4 UTF-8 CRLF {} C++ Go Live
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